

01 DDL, DML, DCL, TCL

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> CREATE DATABASE IF NOT EXISTS practice_db;
Query OK, 1 row affected (0.03 sec)
```

```
mysql> USE practice_db;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->   txn_id INT PRIMARY KEY,
  ->   customer_name VARCHAR(50),
  ->   branch_name VARCHAR(50),
  ->   transaction_type VARCHAR(20),
  ->   amount DECIMAL(10,2),
  ->   transaction_date DATE
  -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> ALTER TABLE bank_transactions
  -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE bank_transactions
  -> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.11 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> RENAME bank_transactions
  -> TO customer_transactions;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'bank_transactions TO customer_transactions' at line 1
mysql> RENAME TABLE bank_transactions TO customer_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> TRUNCATE TABLE bank_transactions;
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
mysql> ^C
mysql> DROP TABLE bank_backup;
ERROR 1051 (42S02): Unknown table 'practice_db.bank_backup'
mysql> INSERT INTO bank_transactions VALUES
  -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
  -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
  -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
  -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
  -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
  -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
  -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
  -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
  -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
  -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
mysql> INSERT INTO customer_transactions
```

```
  -> VALUES
  -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
  -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
  -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
  -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
  -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
  -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
  -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
  -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
  -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
  -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO bank_transactions
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
mysql>
mysql> INSERT INTO customer_transactions
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql> INSERT INTO customer_transactions
  -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
  -> VALUES
  -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE customer_transactions
  -> SET amount = 5000
  -> WHERE txn_id = 105;
Query OK, 1 row affected (0.01 sec)
```

```
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> DELETE FROM customer_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM customer_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM customer_transactions
-> WHERE transaction_type='Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM customer_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Anil	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	NULL
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

```
10 rows in set (0.00 sec)
```

```
mysql> GRANT SELECT
-> ON customer_transactions
-> TO Auditor1;
```

```
ERROR 1410 (42000): You are not allowed to create a user with GRANT
```

```
mysql> SELECT
-> ON customer_transactions
-> TO Auditor1;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ON customer_transactions TO Auditor1' at line 2
```

```
mysql>
```

```
mysql> REVOKE SELECT
```

```
-> ON bank_transactions
```

```
-> FROM Auditor1;
```

```
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
```

```
mysql>
```

```
mysql> UPDATE bank_transactions
```

```
-> SET amount=6000
```

```
-> WHERE txn_id=101;
```

```
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
```

```
mysql>
```

```
mysql> COMMIT;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> SAVEPOINT Before_Update;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE customer_transactions
```

```
-> SET amount=99999
```

```
-> WHERE txn_id=102;
```

```

Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE bank_transactions
    -> SET amount=7000
    -> WHERE txn_id=101;
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount=9000
    -> WHERE txn_id=102;
ERROR 1146 (42S02): Table 'practice_db.bank_transactions' doesn't exist
mysql>
mysql> ROLLBACK TO SP1;
ERROR 1305 (42000): SAVEPOINT SP1 does not exist
mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

```

```

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
+-----+-----+-----+-----+-----+-----+-----+

```

```

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

```